

MARK SCHEME for the October/November 2007 question paper

9709/04

9709 MATHEMATICS

Paper 4, maximum raw mark 50

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- CIE will not enter into discussions or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the October/November 2007 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.



Page 2	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2007	9709	04

Mark Scheme Notes

Marks are of the following three types:

- M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.
 - A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).
 - B Mark for a correct result or statement independent of method marks.
- When a part of a question has two or more "method" steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
 - The symbol $\checkmark$ implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously "correct" answers or results obtained from incorrect working.
 - Note: B2 or A2 means that the candidate can earn 2 or 0.
B2/1/0 means that the candidate can earn anything from 0 to 2.

The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.
- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking g equal to 9.8 or 9.81 instead of 10.

Page 3	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2007	9709	04

The following abbreviations may be used in a mark scheme or used on the scripts:

AEF	Any Equivalent Form (of answer is equally acceptable)
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
BOD	Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)
CAO	Correct Answer Only (emphasising that no "follow through" from a previous error is allowed)
CWO	Correct Working Only - often written by a 'fortuitous' answer
ISW	Ignore Subsequent Working
MR	Misread
PA	Premature Approximation (resulting in basically correct work that is insufficiently accurate)
SOS	See Other Solution (the candidate makes a better attempt at the same question)
SR	Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

Penalties

MR -1	A penalty of MR -1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become "follow through $\sqrt{}$ " marks. MR is not applied when the candidate misreads his own figures - this is regarded as an error in accuracy. An MR -2 penalty may be applied in particular cases if agreed at the coordination meeting.
PA -1	This is deducted from A or B marks in the case of premature approximation. The PA -1 penalty is usually discussed at the meeting.

Page 4	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2007	9709	04

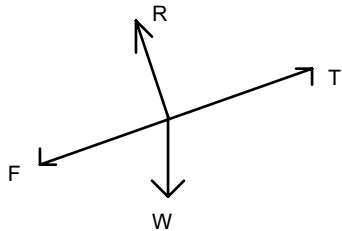
1	DF – 550 = 900x0.2	M1	For using Newton's second law (3 terms)
	[P = 730x30 ÷ 1000]	A1	
	P = 21.9	M1	For using P = (DF)v
		A1	4
2		M1	For applying $s = ut + \frac{1}{2}at^2$ or $(u + at)^2 = u^2 + 2as$ with $a = \pm g$ (either particle)
	$s_1 = 12.5t - \frac{1}{2}gt^2$, $s_2 = \pm \frac{1}{2}gt^2$ or	A1	
	$(12.5 - gt)^2 = 12.5^2 - 2gs_1$ and		
	$(gt)^2 = 2gs_2$		
	[$12.5t - \frac{1}{2}gt^2 + \frac{1}{2}gt^2 = 10$]	M1	For using $s_1 + s_2 = 10$
	t = 0.8s	A1	
	or		
	$2s_1 = 25\sqrt{2 - 0.2s_1} - (20 - 2s_1)$ (or better)		
	Height is 6.8m	A1ft	5 ft for $12.5t - 5t^2$ or $10 - 5t^2$ with candidate's t (requires both M marks)
3	(i) [7 = Fcos θ and 4 = Fsin $\theta \rightarrow$	M1	For stating $F^2 = 7^2 + 4^2$ directly or for resolving in the i and j directions and eliminating θ or F
	$F^2 = 7^2 + 4^2$ (or tan $\theta = 4/7$)]		
	F = 8.06	A1	Allow 8.07 from $4 \div \sin 29.7^\circ$
	[7 = 8.06cos θ or 4 = 8.06sin θ]	M1	For stating tan $\theta = 4/7$ directly or for substituting for F or for θ into 7 = Fcos θ or 4 = Fsin θ
	(or 7 = Fcos29.7° or 4 = Fsin29.7°)		
	$\theta = 29.7$	A1	4 Allow 29.8 from $\sin^{-1}(4 \div 8.06)$
			SR for candidates who mix sine and cosine (max 3/4)
			Fsin $\theta = 7$, Fcos $\theta = 4 \rightarrow F^2 = 7^2 + 4^2$ M1
			For tan $\theta = 7/4$ M1
			For F = 7 and $\theta = 60.3^\circ$ A1
(ii)	Magnitude 7 N	B1	
	Direction opposite to that of the force of magnitude 7 N	B1	2 Any equivalent form
4	(i)	M1	For using KE = $\frac{1}{2}mv^2$
	[$\frac{1}{2}mv^2 - \frac{1}{2}m7^2 = mgx5$]	M1	For equation from KE gain = PE loss (3 terms)
	Speed is 12.2ms ⁻¹	A1	3
			SR for candidates who treat AB as straight and vertical (max 1mark out of 3)
			$v^2 = 7^2 + 2g5 \rightarrow v = 12.2$ B1
	(ii)	M1	For using WD = PE loss – KE gain or WD = KE at B in (i) – actual KE at B
	WD = 0.35x10x5 – $\frac{1}{2}0.35(11^2 - 7^2)$ or	A1ft	ft wrong v in part (i) or for 12.2 scored by B1 in (i)
	WD = $\frac{1}{2}0.35(12.2^2 - 11^2)$		
	Work done is 4.9 J	A1	3 This mark is not available if v = 12.2 is used, having been scored by B1 in part (i)
			SR for candidates who treat AB as straight and vertical, and resistance as constant (max 1mark out of 3)
			a = 7.2 ms ⁻² , R = 0.98 N, WD = 4.9 J B1
			SR for candidates who write 'Resistance =' instead of 'WD =' (max 2/3)
			$0.35x10x5 - \frac{1}{2}0.35(11^2 - 7^2)$ or
			$\frac{1}{2}0.35(12.2^2 - 11^2)$ seen B1
			Answer 4.9J (NB J seen) B1

Page 5	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2007	9709	04

5	(i)	M1		For resolving horizontally (normal force must have a horizontal component)
	$R = T \sin 60^\circ$	A1		
	$[F = W + T \cos 60^\circ]$	M1		For resolving vertically (allow if normal force is not horizontal but equation must contain F, W and T)
	$F = 40 + T \cos 60^\circ$	A1ft	4	ft – allow $F = 40 + T \sin 60^\circ$ following $R = T \cos 60^\circ$
	(ii)	M1		For using $F = \mu R$
	$40 + 0.5T = 0.7 \times 0.866T$	A1ft		Any correct form
	$T = 377$	A1	3	ft unsimplified with candidate's F(T) (with 2 terms) and R(T)

6	(i)	M1		For using $v = u + at$ (or $v^2 = u^2 + 2as$)
	Time taken is 100s	A1		
	(or Distance is 200 m)			
	Distance is 200 m	A1ft	3	ft $s = 2t$ or $1.8t + 0.002t^2$ (or $t = s/2$)
	(or Time taken is 100s)			
	(ii) (a)	M1		For integrating $v(t)$ to find $s(t)$
	$s = k(100t^2 - t^3/3) (+C)$	A1		
	$[k(100 \times 100^2 - 100^3/3) = 200]$	DM1		For using $s(0) = 0$ (may be implied) and $s(100) = 200$
	$k = 0.0003$	A1	4	
	(b) Speed is 3 ms^{-1}	B1ft	1	ft candidate's t and/or k .
	(iii)	M1		For straight line segment, $v(t)$ +ve and increasing throughout (including at $t = 0$)
		M1		For parabolic segment through origin, with +ve slope
	Parabolic segment has decreasing slope; sketches correct relative to each other (line crosses curve once)	A1	3	Depends on both M marks

Page 6	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2007	9709	04

7	(i)		B1	1	The components F and R may be represented by a single contact force, which must be shown at an acute angle to the downward slope.
	(ii)		M1		For finding the resultant upward force (RUF) (3 terms required)
		$T - F - 0.13g$ (16/65)	A1		
		$[T - F - 0.13g$ (16/65) $> 0]$	M1		For use of $RUF > 0$ (since P starts to move upwards).
		$T - F > 0.32$	A1	4	AG
	(iii)	$R = 0.13g(63/65)$ or $0.13g \cos 14.25\dots$ (= 1.26)	B1ft		ft 0.13g cos 75.7.....
		$F = 0.6 \times 1.26$ (= 0.756)	M1		For using $F = \mu R$
			M1		For applying Newton's second law to P (4 terms required) or to Q (3 terms required)
					or
		$T - F - 0.32 = 0.13a$ and $0.11g - T = 0.11a$	A1ft		for using $W_Q - W_P \sin \alpha - F = (m_P + m_Q)a$ ft 1.26 instead of 0.32 following a consistent sin/cos mix throughout (i) and (ii)
		or $0.11g - F - 0.32 = (0.13 + 0.11)a$			
		Acceleration is 0.1 ms^{-2}	M1		For substituting for F and solving for a.
			A1	6	